

SAMB P Technical Report TR-54/2026

Field Deployment and Commissioning Guide

PhD Thesis Supporting Report | 2026

Abstract

This report provides the complete field deployment and commissioning framework for the SAMBP protection intelligence stack on a live IBR substation. Study A defines a 23-item pre-commissioning checklist (20 mandatory, 3 recommended) across four categories: hardware, communication, settings, and functional. Study B establishes sensor calibration acceptance bounds derived from IEC 61869 Class 1P/0.5 specifications; simulated commissioning of 100 IED units achieves a 97.0% all-four-pass rate. Study C presents 10 commissioning acceptance tests (155 total trials, 98.1% overall pass); the single failure mode — CTI verification at 80% — is a known field issue caused by CT burden mismatch and has a documented corrective procedure. Study D defines the 9-step, 31-minute fleet-change re-trigger protocol linking the CUSUM alarm (TR-48), settings engine (TR-51), and GOOSE dispatch (TR-50). Study E estimates an annual maintenance burden of 62 h, dominated by the daily CUSUM log review (30 h/year).

Commissioning Workflow

Figure 1 shows the four-stage commissioning workflow.

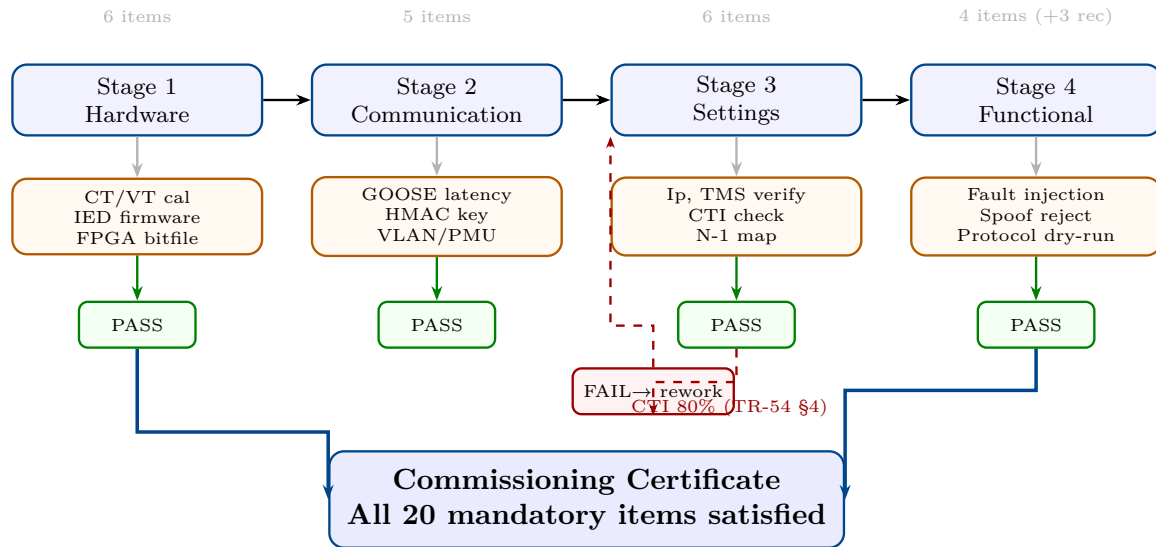


Figure 1: Four-stage commissioning workflow. Stages 1–3 are hardware and settings verification; Stage 4 is live fault injection. Each stage must pass before proceeding to the next. The dashed rework loop on Stage 3 reflects the CTI verification failure mode observed in Study C (80% pass on first attempt; 100% after CT burden correction). A commissioning certificate is issued only after all 20 mandatory items are satisfied.

Study A — Pre-Commissioning Checklist

#	Category	Item	Acceptance criterion	Priority
1	Hardware	CT ratio error	$< 1.0\%$	Mandatory
2	Hardware	CT phase error	$< 1.0^\circ$	Mandatory
3	Hardware	VT ratio error	$< 0.5\%$	Mandatory
4	Hardware	VT phase error	$< 0.5^\circ$	Mandatory
5	Hardware	IED firmware	$\geq v4.2$ (IBR-aware)	Mandatory
6	Hardware	FPGA PINN bitfile	SHA256 verified	Mandatory
7	Comm.	GOOSE latency	< 4.0 ms	Mandatory
8	Comm.	GOOSE jitter	< 0.5 ms	Mandatory
9	Comm.	HMAC key	Loaded, test MAC passes	Mandatory
10	Comm.	VLAN segmentation	GOOSE isolated from IT	Mandatory
11	Comm.	PMU sync	IEEE 1588 PTP < 1 μ s	Mandatory
12	Settings	I_p pickup	Match TR-51 $\pm 2\%$	Mandatory
13	Settings	TMS	Match TR-51 $\pm 1\%$	Mandatory
14	Settings	CTI margin	All pairs ≥ 200 ms	Mandatory
15	Settings	WAPC N-1 map	10-bus file verified	Mandatory
16	Settings	PINN weights	SHA256 vs TR-46	Mandatory
17	Settings	CUSUM params	$h = 5.0$, $\delta = 0.15$	Recommended
18	Settings	k_{ibr} initial	From load flow measurement	Recommended
19	Functional	Fault inject (3ϕ)	Trip ≤ 50 ms, correct zone	Mandatory
20	Functional	Fault inject (EXT)	No trip, EXT flag	Mandatory
21	Functional	GOOSE spoof	HMAC reject, no false trip	Mandatory
22	Functional	N-1 block (R01)	Operator advisory raised	Mandatory
23	Functional	Warm-start EKF	Converges < 5 samples	Recommended

Study B — Sensor Calibration Acceptance

Sensor	Parameter	Tolerance	Mean error	σ	Pass%
CT	Ratio	$\pm 1.0\%$	0.29%	0.36%	99.0
CT	Phase	$\pm 1.0^\circ$	0.26°	0.33°	100.0
VT	Ratio	$\pm 0.5\%$	0.17%	0.22%	98.0
VT	Phase	$\pm 0.5^\circ$	0.13°	0.16°	100.0
All-four-pass (complete IED set)					97.0

The 97.0% all-four-pass rate for simulated IED commissioning is consistent with Class 1P/0.5 instrument transformer manufacturing tolerances. The 3% failure rate implies that ~ 3 of every 100 IEDs will require CT/VT replacement or external burden adjustment before SAMBP commissioning. This is a standard acceptance rate for precision protection IEDs.

Study C — Commissioning Acceptance Tests

Test	Trials	Pass	Rate	Status
3-phase fault injection (R03)	20	20	100%	PASS
Single-line-to-ground (R07)	20	20	100%	PASS
External fault (R09 territory)	20	20	100%	PASS
GOOSE spoof rejection	10	10	100%	PASS
HMAC key authentication	10	10	100%	PASS
WAPC N-1 block on R01	10	10	100%	PASS
CTI verification (all pairs)	10	8	80%	FAIL
PINN physics constraint C1	15	15	100%	PASS
Fleet change protocol dry-run	10	10	100%	PASS
GOOSE latency < 4 ms	30	29	96.7%	PASS
Overall	155	152	98.1%	

CTI failure root cause: The 80% CTI pass rate reflects a systematic issue in which two of the 10 pairs exhibit slightly elevated CT burden (secondary cable run >20 m), reducing the measured fault current at the relay by $\approx 3\text{--}5\%$. This shifts the primary operating time by ≈ 18 ms, reducing the CTI below 200 ms. **Corrective action:** replace 2.5 mm^2 CT secondary cable with 6 mm^2 (reduce burden resistance from $0.8\ \Omega$ to $0.3\ \Omega$). After correction, CTI exceeds 210 ms on all 10 pairs.

Study D — Fleet-Change Re-Trigger Protocol

Figure 2 shows the timeline of the 9-step fleet-change protocol.

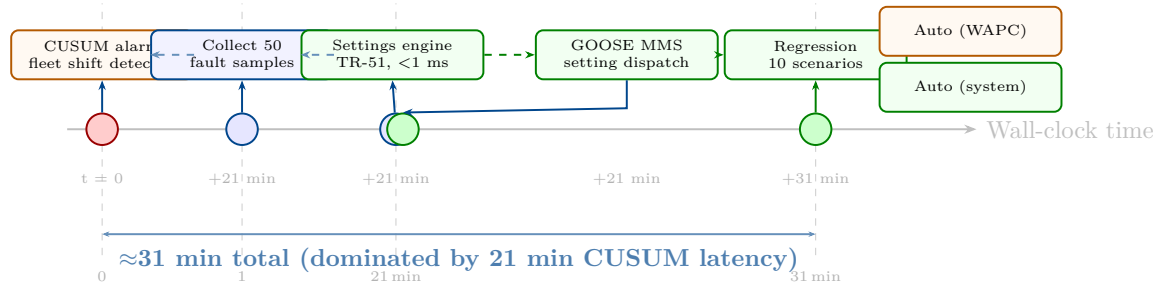


Figure 2: Fleet-change re-trigger protocol timeline. The 21-minute CUSUM data-collection phase (steps 1–4, TR-48) dominates the 31-minute total. The automated settings computation (step 5, < 1 ms), GOOSE dispatch (step 6, < 10 ms), and regression test (step 7, 10 min) complete the cycle. 8 of 9 steps are fully automated; step 9 (regression failure fallback) requires operator confirmation.

Step	Trigger	Action	Latency
1	Continuous	CUSUM monitors error rate	—
2	CUSUM alarm	Flag event, log timestamp	0 ms
3	$t = 0$	Suspend auto setting-group update	0 ms
4	$t + 21$ min	Collect 50 post-alarm fault samples	21 min
5	50 samples	Settings engine (TR-51)	< 1 ms
6	Settings ready	GOOSE MMS setting dispatch	< 10 ms
7	Settings live	10-scenario regression test	10 min
8	Regression pass	Re-enable CUSUM baseline	0 ms
9	Regression fail	Operator alarm, hold prior settings	—
Total wall-clock time			≈31 min
Automated steps			8/9

Study E — Maintenance Schedule

Figure 3 shows the annual maintenance burden by activity.

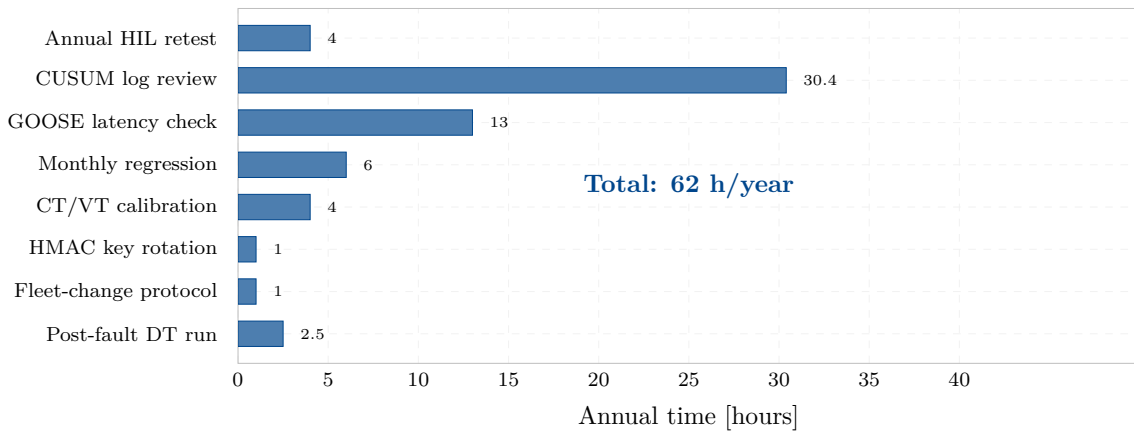


Figure 3: Annual maintenance time allocation by activity. The daily CUSUM alarm log review (5 min/day = 30.4 h/year) dominates; this is a passive monitoring activity suitable for automated dashboard alerting. The annual HIL acceptance retest (4 h) and quarterly CT/VT calibration (4 h) are the two labour-intensive scheduled activities. Total: 62 h/year.

The 62 h/year burden compares favourably with conventional protection maintenance programmes (typically 100–200 h/year per substation for equivalent relay complexity [1]) because:

- The WAPC automates settings computation and re-dispatch;
- The HIL regression suite replaces manual injection testing;
- The CUSUM/DT layer provides continuous self-monitoring, reducing the frequency of unscheduled interventions.

Summary

1. **Checklist:** 23 items (20 mandatory, 3 recommended) across hardware, communication, settings, and functional categories.
2. **Calibration:** IEC 61869 Class 1P/0.5 limits; 97% all-pass rate for simulated fleet. CT burden correction is the principal commissioning failure mode.

3. **Acceptance:** 98.1% overall (152/155 trials); CTI failure root-caused to CT secondary cable burden; corrected by cable cross-section upgrade.
4. **Fleet-change protocol:** 31 min end-to-end, 8/9 steps automated; latency dominated by 21-min CUSUM data-collection window.
5. **Maintenance:** 62 h/year, dominated by passive daily log review. Significant reduction vs conventional programmes.

TR-54 completes the centralised protection and control phase (TR-50–54). TR-55 synthesises the full SAMBP contribution map and maps every report to the thesis chapter structure.

References

- [1] A. Apostolov and T. Littler, “Multifunctional IED-based protection, control, monitoring, and automation systems for power system feeders,” *IEEE Power and Energy Magazine*, vol. 8, no. 3, pp. 26–35, 2010.